

Java Programming Practice Problems

Question 1: Percentage Marks Calculator

```
import java.util.Scanner;

public class Q1_Percentage {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter obtained marks: ");
        double obtained = sc.nextDouble();
        System.out.print("Enter maximum marks: ");
        double max = sc.nextDouble();

        double percentage = (obtained / max) * 100;
        System.out.println("Percentage = " + percentage + "%");
    }
}
```

Question 2: Circumference and Area of a Circle

```
import java.util.Scanner;

public class Q2_Circle {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter radius of circle: ");
        double r = sc.nextDouble();

        double pi = 3.1416;
        double circumference = 2 * pi * r;
        double area = pi * r * r;

        System.out.println("Circumference = " + circumference);
        System.out.println("Area = " + area);
    }
}
```

Question 3: Discounted Selling Price

```
import java.util.Scanner;

public class Q3_Discount {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter original selling price: ");
        double price = sc.nextDouble();
        System.out.print("Enter discount percentage: ");
        double discount = sc.nextDouble();

        double discountedPrice = price - (price * discount / 100);
        System.out.println("Discounted selling price = " + discountedPrice);
    }
}
```

Question 4: Original Price from Discounted Price

```
import java.util.Scanner;

public class Q4_OriginalPrice {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter discounted selling price: ");
        double discountedPrice = sc.nextDouble();
        System.out.print("Enter discount percentage: ");
        double discount = sc.nextDouble();

        double originalPrice = discountedPrice / (1 - discount / 100);
        System.out.println("Original selling price = " + originalPrice);
    }
}
```

Question 5: Electrical Power Calculation (Watts = Volts × Amperes)

```
import java.util.Scanner;

public class Q5_Watts {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter voltage in volts: ");
        double volts = sc.nextDouble();
        System.out.print("Enter current in amperes: ");
        double amps = sc.nextDouble();

        double watts = volts * amps;
        System.out.println("Power in Watts = " + watts);
    }
}
```

Question 6: Area of a Trapezoid

```
import java.util.Scanner;

public class Q6_Trapezoid {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter length of parallel side a: ");
        double a = sc.nextDouble();
        System.out.print("Enter length of parallel side b: ");
        double b = sc.nextDouble();
        System.out.print("Enter height h: ");
        double h = sc.nextDouble();

        double area = h * (a + b) / 2;
        System.out.println("Area of trapezoid = " + area);
    }
}
```

Question 7: Even or Odd Checker

```
import java.util.Scanner;

public class Q7_EvenOdd {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter an integer: ");
        int num = sc.nextInt();

        if (num % 2 == 0)
            System.out.println("EVEN");
        else
            System.out.println("ODD");
    }
}
```

Question 8: Area of a Triangle using Heron's Formula

```
import java.util.Scanner;
import java.lang.Math;

public class Q8_Heron {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter side a: ");
        double a = sc.nextDouble();
        System.out.print("Enter side b: ");
        double b = sc.nextDouble();
        System.out.print("Enter side c: ");
        double c = sc.nextDouble();

        double s = (a + b + c) / 2; // semi perimeter
        double area = Math.sqrt(s * (s - a) * (s - b) * (s - c));

        System.out.println("Area of triangle = " + area);
    }
}
```